

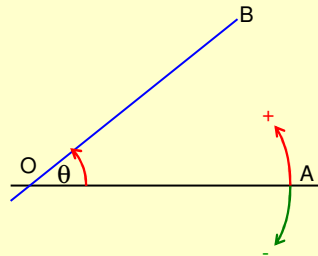


## FSK 116 – Semester 1 Mathematics and Other Essentials

### Priorities

- Know how YOUR calculator works and always have YOUR calculator with you.
- Always have a pencil (and an eraser) at hand when doing Physics.

# Angles

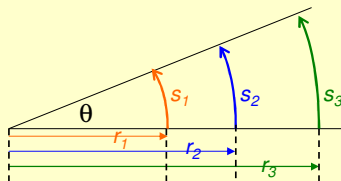


- Angle  $\theta$  is positive for anti-clockwise rotation
- Angle  $\theta$  is negative for clockwise rotation
- 1 Rotation =  $360^\circ$
- $\frac{1}{2}$  Rotation =  $180^\circ$
- $\frac{1}{4}$  Rotation =  $90^\circ$

$$\frac{s_1}{r_1} = \frac{s_2}{r_2} = \frac{s_3}{r_3}$$

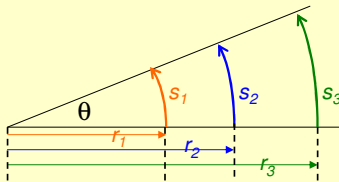
$$\Rightarrow \frac{s}{r} = \theta$$

- Radians (rad)



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## Angles - 2



$$\frac{s_1}{r_1} = \frac{s_2}{r_2} = \frac{s_3}{r_3} = \text{constant}$$

$$\Rightarrow \frac{s}{r} = \theta$$

- 1 Rotation =  $\frac{\text{circ of circle}}{r} = \frac{2\pi}{r}$
- $\therefore$
- 1 Rotation =  $2\pi$  radians ( $360^\circ$ )
- $\frac{1}{2}$  Rotation =  $\frac{1}{2} \frac{\text{circ of circle}}{r} = \frac{1}{2} \frac{2\pi}{r}$   
=  $\pi$  radians ( $180^\circ$ )
- $\frac{1}{4}$  Rotation =  $\frac{1}{4} \frac{\text{circ of circle}}{r} = \frac{1}{4} \frac{2\pi}{r}$   
=  $\pi/2$  radians ( $90^\circ$ )
- $\therefore$
- $1^\circ = 0.01745 \text{ rad}$
- $1 \text{ rad} = 57.3^\circ$

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## Calculator Skills

- Change angle setting between degrees – radian – gradient: **DGR**
- Convert angle between degrees – radian – gradient: **DGR ►**
- Set decimal and exponents: **FSE**
  - Fixed – Scientific – Engineering
  - Decimal –  $10^x$  –  $10^{3n}$
- Using  $10^x$ 
  - $3 \times 10^2$  press 3 then *Exp* then 2
  - *Exp*  $\equiv \times 10$

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## Examples

- |                                                                                                  |                         |
|--------------------------------------------------------------------------------------------------|-------------------------|
| • Convert to Radians                                                                             | • Convert to degrees    |
| – $1^\circ$                                                                                      | – $\frac{1}{2}$ rad     |
| – $32^\circ$                                                                                     | – $\frac{1}{2} \pi$ rad |
| – $175^\circ$                                                                                    | – $\frac{4}{3} \pi$ rad |
| – $270^\circ$                                                                                    | – $\frac{2}{3} \pi$ rad |
| – $315^\circ$                                                                                    | – 7 rad                 |
| • A circle has a radius of 3 cm<br>what is the arc length<br>subtended by an angle of $50^\circ$ | – 6 rad                 |
|                                                                                                  | – 1 rad                 |

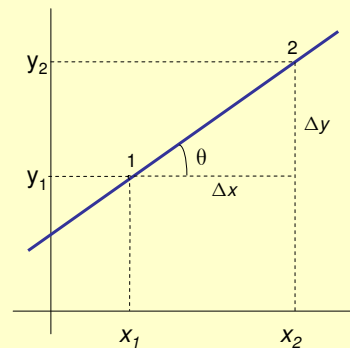
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# Differentiation

## Gradient of a straight line

- $y = mx + c$
- $m, c$  – constants
- Gradient – tempo at which the  $y$  value changes with increasing  $x$ .
  - Gradient =  $\frac{\text{change in } y}{\text{change in } x}$ .
- Co-ordinates of point 1 and point 2.

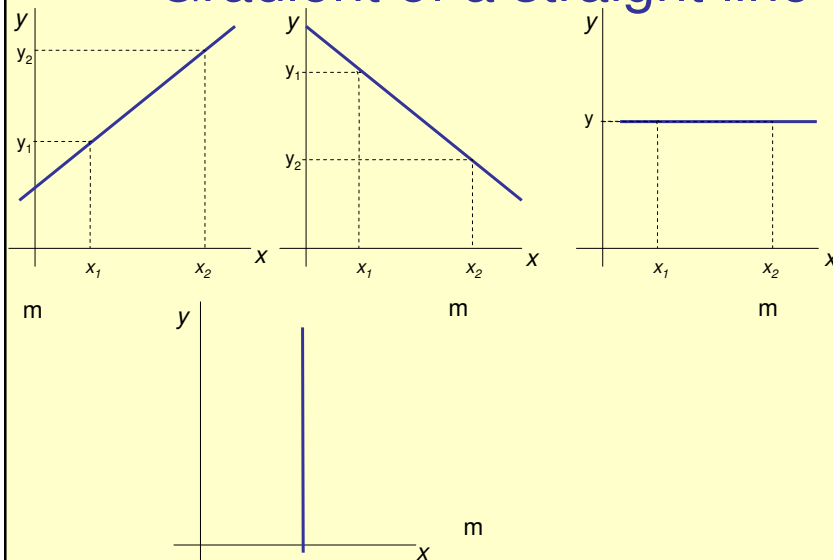


$$\begin{aligned} \text{Gradient} &= \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1} \\ &= \frac{(mx_2 + c) - (mx_1 + c)}{x_2 - x_1} \\ &= m \end{aligned}$$

Also:

$$\tan \theta = \frac{\Delta y}{\Delta x} = \text{gradient}$$

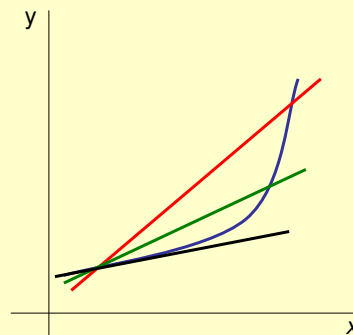
## Gradient of a straight line



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## Gradient of a curve

- Gradient changes at each point
- Also depends on the size of interval  $\Delta x$ .
- Need new def. for gradient
  - The gradient of the tangent to the curve at the point.
- Can determine the gradient of the tangent using previous method.
  - Graphical method – not always successful.



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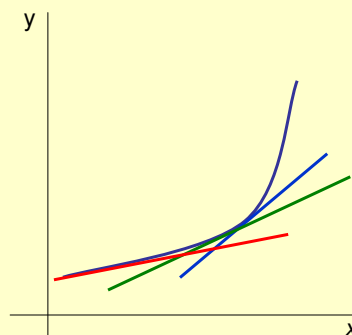
## Gradient of a curve

- Gradient changes at each point i.e it's a function of  $x$
- Can be expressed as a function – the derivative.
- Derivative notations:

$$\frac{d}{dx}(y) \quad \bullet \quad \text{d-d-x of } y$$

$$\frac{dy}{dx} \text{ or } dy/dx \quad \bullet \quad \text{d-y-d-x}$$

$$D(y) \text{ or } D_x y \quad \bullet \quad \text{d of } y \text{ or d-x of } y$$



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## Handy Derivatives

$$y = x^n \quad \text{then} \quad \frac{dy}{dx} = nx^{n-1}$$

$$y = ax^n \quad \text{then} \quad \frac{dy}{dx} = anx^{n-1}$$

$$y = \sin x \quad \text{then} \quad \frac{dy}{dx} = \cos x$$

$$y = \sin ax \quad \text{then} \quad \frac{dy}{dx} = a \cos ax$$

$$y = \cos x \quad \text{then} \quad \frac{dy}{dx} = -\sin x$$

$$y = \cos ax \quad \text{then} \quad \frac{dy}{dx} = -a \sin ax$$

$$y = \ln x \quad \text{then} \quad \frac{dy}{dx} = \frac{1}{x}$$

$$y = a^x \quad \text{then} \quad \frac{dy}{dx} = (\ln a) a^x$$

$$y = e^x \quad \text{then} \quad \frac{dy}{dx} = e^x$$

$$y = e^{f(x)} \quad \text{then} \quad \frac{dy}{dx} = f'(x) e^{f(x)}$$

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# Differentiation Rules

- Sum Rule

$$y = f(x) + g(x)$$

$$\frac{dy}{dx} = \frac{df}{dx} + \frac{dg}{dx}$$

$$y = (2x^2 + x + 3) + (3x^2 + 2x + 1)$$

- Product Rule

$$y = f(x)g(x)$$

$$\frac{dy}{dx} = f \frac{dg}{dx} + g \frac{df}{dx}$$

$$y = (3x^2 + 5x)(2x^3 - 3)$$

- Quotient Rule

$$y = \frac{f(x)}{g(x)}$$

$$\frac{dy}{dx} = \frac{g \left( \frac{df}{dx} \right) - f \left( \frac{dg}{dx} \right)}{g^2}$$

$$y = \frac{9x^7}{(x^2 + 1)}$$

- Chain Rule

$$y = y(u), \text{ and } u = u(x)$$

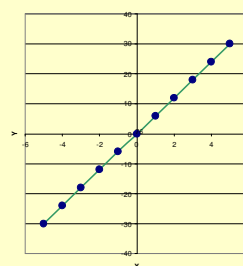
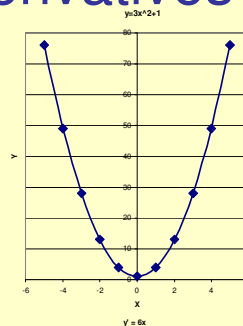
$$\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx}$$

$$y = \sin(4x^2)$$

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# Higher order Derivatives

- If  $y = 3x^2 + 1$
- Then  $\frac{dy}{dx} = 6x$
- $dy/dx$  is gradient of  $y$
- Where  $dy/dx = 0$  - a turning point.
- $\frac{d^2y}{dx^2} = 6$
- If  $d^2y/dx^2 < 0$  where  $dy/dx = 0$  - local maximum
- If  $d^2y/dx^2 > 0$  where  $dy/dx = 0$  - local minimum



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## Problems

- Determine the following:

- $\frac{d^2y}{dx^2}$  if  $y = 4x^4 + 2x^2 - 5x - 2$

- $\frac{d^3y}{dx^3}$  if  $y = e^{x^2}$

- $\frac{d^2y}{dx^2}$  if  $y = \frac{x}{x+1}$

- $\frac{d^3y}{dx^3}$  if  $y = 2\sin 3x$

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## Example - 1

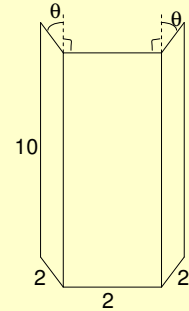
- An object is projected vertically upwards and the height above ground,  $h$  is given by  $h = 20t - 5t^2$  meter, in which the time,  $t$ , is measured in seconds. Calculate the maximum height which it will reach and when it will be reached.

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## Example - 2

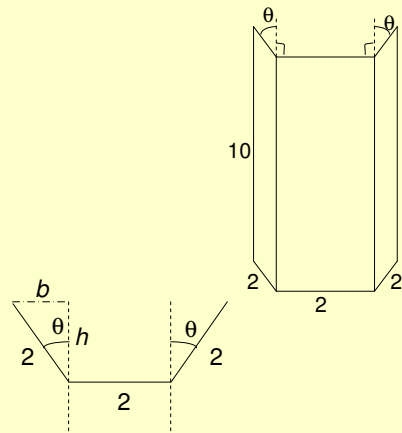
- A flat metal plate (dimensions in cm) is bent in the form of a gutter (see figure below). Find a **formula** for the volume of the gutter in terms of the angle  $\theta$ .



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## Example - 2

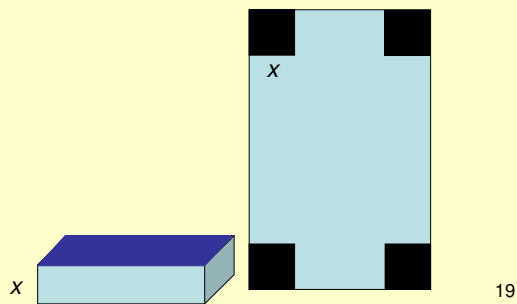
- The volume of a gutter was determined i.t.o. the angle of the sides w.r.t. the bottom. Determine the angle at which the gutter will contain a maximum volume.



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## Example - 3

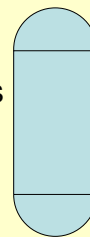
- A farmer possesses a rectangular sheet of polyethylene plastic of 24 m by 9 m which he wishes to use as a waterproof lining for a rectangular dam with constant depth and vertical walls. To achieve this, equal squares of side length  $x$  are cut from the corners and the edges are glued together. Calculate the value of  $x$  for which the volume will be a maximum.



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## Problems

- A sports field must be designed so that the circumference is 440 m. The shape of the field is 2 semi-circles with a rectangular portion in between. Determine the measurements so that the area of the rectangular portion is a maximum.



- Determine the were the maximum and minimum values lie for the following function:

$$y = x^3 - 3x^2 - 24x + 1$$

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